



Immediate Early and Delayed Implants: A 2-Year Prospective Cohort Study of 131 Transmucosal Flapless Implants Placed in Sites With Different Pre-extractive Endodontic Infections

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Marginal bone loss (MBL) is a radiographic parameter to determine bone variation around the implant and still offers some information on dental implant status, osseointegration success, and probably failure risk. A number of conditions may affect the periimplant marginal bone level^{1–5} and among them the presence of endodontic lesions.^{1,6–12} Again, bone quality and implant placement timing influence bone formation and bone loss around implants.¹³

Acute endodontic infections, such as acute periapical lesion/abscess, represent a concrete contraindication to immediate

Introduction: To evaluate clinical outcome of early, immediate, and delayed transmucosal implants placed in patients affected by acute/chronic endodontic lesions.

Materials and Methods: Eighty-five consecutive patients received 131 titanium implants with zirconium-oxide blasted surface. Pre-extractive diagnosis represented the main criteria for implant placement timing, following “best treatment” criteria. Implants were placed with flapless transmucosal technique. Hopeless teeth with chronic periapical lesions received atraumatic extraction, and an implant was immediately placed (Immediate Group, $n = 29$). Teeth with acute periapical lesion/abscess were extracted and implants placed after 8 to 12 weeks (Early Group, $n = 29$). Implants placed 10- to 12-month after extraction constituted the control group (delayed group, $n = 73$). Implants were loaded 3 months after insertion with provisional resin crowns and after approximately 15 days with definitive ceramic crowns. Marginal bone loss

(MBL) was measured in a single-blind manner on periapical radiographs at 1, 3, 6, 12, and 24 months.

Results: Multilevel analysis described exploring factors associated with MBL. Survival rate was 100%. MBL after 24 months was 0.78 ± 0.70 (95% confidence interval [CI]: 0.20 to -1.37) at immediate, 0.48 ± 0.70 at early (95% CI: -0.006 to -0.961), and 1.02 ± 1.01 (95% CI: 0.61 to -1.43) at delayed groups. Implant groups (immediate/early/delayed) and location (maxillary/mandibular) showed statistically significant results. Early group showed the lowest MBL values. The immediate group demonstrated less MBL than the delayed group.

Conclusion: Early implant placement technique preserves periimplant marginal bone level more than immediate and delayed techniques. (Implant Dent 2017;26:654–663)

Key Words: MBL, dental implants, flapless surgery, best clinical practice, chronic periapical lesions

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placement of implants.^{1,6–12,14–18} Waiting for a healed bone site is a possible approach,¹⁶ thus eliminating the

risk of anaerobic bacteria presence in the bone niches and periapical tissue.^{18–20}

However, chronic periapical lesions, deep carious lesions with the absence of acute infections and the presence of intact cortical bone lead to extraction and immediate implant placement, as proposed by many previous investigations.^{21–23} Therefore, clinical decision making may be represented by an immediate implant placement only when no acute endodontic infection/abscess is present at the time of extraction. Otherwise, a placement after 2 to 4 months from tooth extraction (infection resolution with socket tissue healing) may be performed (early implant placement) in a healing new bone.²¹

Different decision-making strategies mean placing implants in extremely different bone conditions (ie, presence/absence of cortical bone, healing bone etc...), that influence periimplant marginal bone remodeling and resorption, suggesting that many questions remain unanswered.^{15,24}

The aim of this consecutive non-randomized prospective cohort study was to evaluate the failure rate, complications, and MBL of implants placed in endodontically affected bone sockets. MBL was evaluated at 1, 3, 6, 12, and 24 months. The decision to immediately replace or to wait for bone healing and place implants after 2 to 3 months (early placement) or after 10 to 12 months (delayed placement) was taken according to a strict predefined clinical protocol, after an accurate evaluation of tooth and periapical status, clinical and radiographic conditions, and assessing the presence/absence of endodontic acute periapical lesion, to achieve the “best clinical practice”.²⁵

MATERIALS AND METHODS

Study Setting and Patient Selection

The study design was a single-blind human longitudinal prospective cohort study comparing the clinical and radiographic outcomes after 2 years of treatment for patients who had been lost one or more teeth for endodontic, traumatic, and deep carious lesions.

The study was conducted in 1 University Endodontic Clinical Department and in 2 private dental offices between January 2011 and January 2017 by the same clinical team.

All patients included in this investigation were treated according to the principles established by the Declaration of Helsinki as modified in 2013.²⁶ Before enrollment, written and verbal information were given by the clinical staff, and each patient gave a written consent according to the above-mentioned principles. An additional signed informed consent was obtained from all patients stating that they accepted the treatment plan and agreed to cover the costs and follow the maintenance hygiene program. This report was written according to the Consolidated Standards of Reporting trials guidelines for reporting clinical trials (STROBE)²⁷ and respecting the guidelines published by Dodson in 2007.²⁸

The patients were considered eligible or noneligible for inclusion in the clinical protocol based on the following criteria:

Inclusion Criteria

- 18 to 75 years of age
- Partially dentate requiring dental implants
- Possibility to be included in a hygiene recall program and implant control for at least 2 years
- Smoking less than 20 cigarettes per day

Exclusion Criteria

- Medical and/or general contraindications for the surgical procedures (American Surgical Association score ≥ 3)
- Poor oral hygiene and lack of motivation
- Active clinical periodontal disease in the dentition expressed by probing pocket depth >4 mm and bleeding on probing
- Smoking more than 20 cigarettes per day
- Uncontrolled diabetes mellitus
- Systemic or local diseases that could compromise postoperative healing and osseointegration
- Alcohol and/or drug abuse
- Pregnancy or lactating
- Malocclusion and other occlusal disorder (bruxism)

• Bisphosphonate therapy

The type of periapical and root lesion represented the main criteria to determine the clinical approach and surgery timing.

Clinical evaluations of periapical status were made by 3 experienced operators who are included as authors.

Choice of the surgical approach and timing of implant placement, (immediate, early, and delayed according to the timing classification proposed by the Third ITI Consensus Conference),¹³ were not determined randomly as the purpose of the study was to use well-defined clinical parameters for the “best clinical practice.”²⁵ Therefore, the choice of the different surgical approach and the consequent clinical decision and implant placement timing (immediate, early, and delayed) were made on the basis of the following rigorous clinical criteria: the presence of acute endodontic periapical lesion (with pain, fistula, exudate/pus, tenderness and radiographic apical translucency, or all of them) and/or the presence of chronic periapical disease (Periapical Index or PAI 3–4).²³

The 3 groups were defined as follows (Fig. 1, Table 1):

- Immediate postextraction implant group (type 1 for ITI)¹⁴: When the implant was placed into a fresh extraction socket immediately after extraction of the root affected by chronic periapical disease and/or seriously damaged hopeless teeth. Only chronic periapical lesions were present and identified by periapical radiolucency.
- Early implant group (type 2 for ITI)¹³: When the implant was placed in healed bone after 8 to 12 weeks after extraction of root affected by acute periapical lesion and/or abscess, pus, and clinical symptoms.
- Delayed implant group (type 4 for ITI)¹³: When the implant was placed in edentulous mature bone 10 to 12 months after the tooth extraction.

Surgical Procedures

Cylindrical implants (SP Premium; Sweden & Martina, Padova,

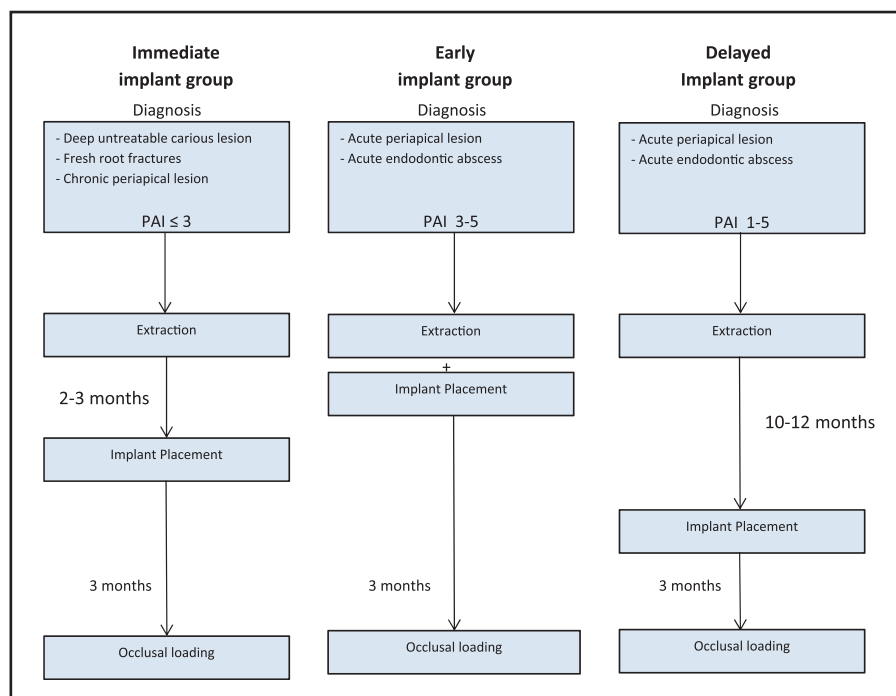


Fig. 1. Flowchart of the study. Clinical and radiological diagnosis determined the choice of the different surgical approach and the consequent timing of implant placement. Periapical Index (PAI) was used to evaluate the presence/absence of endodontic periapical lesion of the extracted tooth. All implants were loaded approximately three months after implant insertion.

Italy) with zirconium-oxide blasted (ZirTi) surface, smooth machined collar 0.5 mm, switching platform emergence profile 0.3 mm (tulip-shape profile), and hexagonal internal connection 3.5 or 4.25 or 5.0 mm diameter were used.

One single experienced surgeon performed all surgeries. Before surgery, a careful occlusal and periodontal examination was performed

on each patient, including presence of plaque, gingivitis, pocket depth, and radiographic bone loss of all remaining teeth. Oral hygiene instruction and periodontal therapy were performed when and where indicated.

Two days before the intervention, all patients were asked to comply with a pharmacological regime that included amoxicillin/clavulanic acid

1 gr tablet and application of chlorhexidine digluconate 0.20% gel (Corsodyl Gel; GlaxoSmithKline UK, Brentford, United Kingdom) twice a day. Antibiotic administration continued during 5 to 6 days after surgery.

All surgical procedures were conducted under local anesthesia with mepivacaine chlorohydrate 30 mg/mL (Carboplyina; Dentsply Italia srl, Roma, Italy). The edge of the implant platform was placed at the margin of the buccal bone level. The torque value was 20 to 70 N/cm². In all implants, an adequate primary stability was obtained. Implants were placed to obtain transmucosal not submerged healing and to avoid a second stage surgery to expose screw. A 1-mm high cover screw was applied. No computer-aided guide was used.

Immediate Implant Placement

For immediate postextractive insertion (Immediate Group), an atraumatic flapless root extraction was performed, and a careful inspection of the socket site was made. All granulation tissue were gently debrided from the apical portion of the socket.

Then a 1.2-mm drill was used to prepare the intrasocket place, following the palatal bony walls as a guide. Twist and calibrated drills at 225 rpm were then used and irrigated with sterile saline solution.

Primary implant stability was obtained by anchoring the implant in the remaining apical portion of the socket at least 3 mm beyond the root apex area.

A porcine corticocancellous bone substitute (Osteobiol MP3; Tecknoss Dental, Coazze, Italy) was applied into the surgical site to fill the socket and to reduce any gaps between the implant and the residual bone.

The cover screw resulted exposed and positioned, just over the gingival level of higher crestal bone level.

Early and Delayed Implant Placement

The surgical procedures were similar for the early and delayed groups. No flaps were reflected, and no computer-aided surgical guides were used.

Table 1. Time of implant placement according to the clinical diagnosis

Groups	Time of implant placement	Number of implants	Reason for extraction
Immediate	Immediate post-extraction	29	Deep untreatable carious lesions Fresh root fractures Chronic periapical lesions
Early	8-12 wk	29	Acute periapical lesions Acute endodontic abscess
Delayed	12 mo	73	Acute Endodontic abscess Root fractures Chronic periapical lesion Deep untreatable carious lesions

Of 131 implants inserted, 29 implants were placed immediately into fresh extraction sockets, 29 implants were placed in healed bone after 8 to 12 weeks from teeth extraction, and 73 implants were placed in 10 to 12 months edentulous areas.

Table 2. Distribution and Number of the Placed Implants (85 Patients)

Implant location	Mandible	54
	Maxilla	77
Implant placement	Immediate	29
	Early	29
	Delayed	73
Total		131

A total of 85 patients fulfilled inclusion criteria. Implant groups were determined on the basis of implant placement group (immediate/early/delayed) and implant location (maxilla/mandible).

An initial 1.2-mm diameter drill was used to mark the position, angle, and depth. The drill passed through the mucosa (transmucosal), cortical bone, and cancellous bone under copious saline irrigation. A twist and calibrated drill at 225 rpm was used, and a site of the adequate depth and diameter was created while irrigating with sterile saline solution.

In all cases, the entire rough surface region of the implant was positioned approximately 1 mm under the cortical bone level and neck approximately at the mucosa level. The cover screw resulted exposed, just over the gingival surface.

Postoperative Procedures

A surgical dressing (Coe-Pak; GC, Tokyo, Japan) was placed on the wound in all patients. One week after the implant surgery, the dressing was removed and the first clinical control was performed.

Patients were instructed to follow a soft diet regime for 1 week, to rinse 3 times/d with 0.12% chlorhexidine mouthwash for 3 weeks, and to perform oral hygiene on the Coe-Pak using a normal-medium toothbrush for the first week and for 2 weeks after surgical pack removal. Thereafter, conventional brushing and flossing were permitted.

Prosthetic Rehabilitation

Three months later, an impression was taken using the pick-up technique with custom trays for analogs technique, and the customized titanium abutment was fixed within 5 to 7 days.

An acrylic temporary single crown was cemented with provisional cement (Temp Bond; Kerr, Scafati,

Italy) as a provisional prosthetic rehabilitation.

Twelve to 15 days later, a definitive prosthetic metal-ceramic rehabilitation, made by 2 equally experienced prosthodontists, were positioned and fixed with polycarboxylate cement (Heraeus Kulzer GmbH, Hanau, Germany).

Follow-up Implant Evaluation

Active periodontal therapy consisting of motivation, instruction in oral hygiene practice, scaling, and root planning was performed until no periodontal (modest) disease was present, no bleeding on probing and pocket probing depth ≥ 3 mm were detected during the follow-up procedures.

Radiographic Assessment

Intraoral periapical radiographs of all implants were taken using a paralleling technique with Rinn-holders and analog films (Kodak Ektaspeed Plus; Eastman Kodak Co., Rochester, NY) after implant placement (baseline) and at 1, 3, 6, 12, and 24 months after implant insertion.

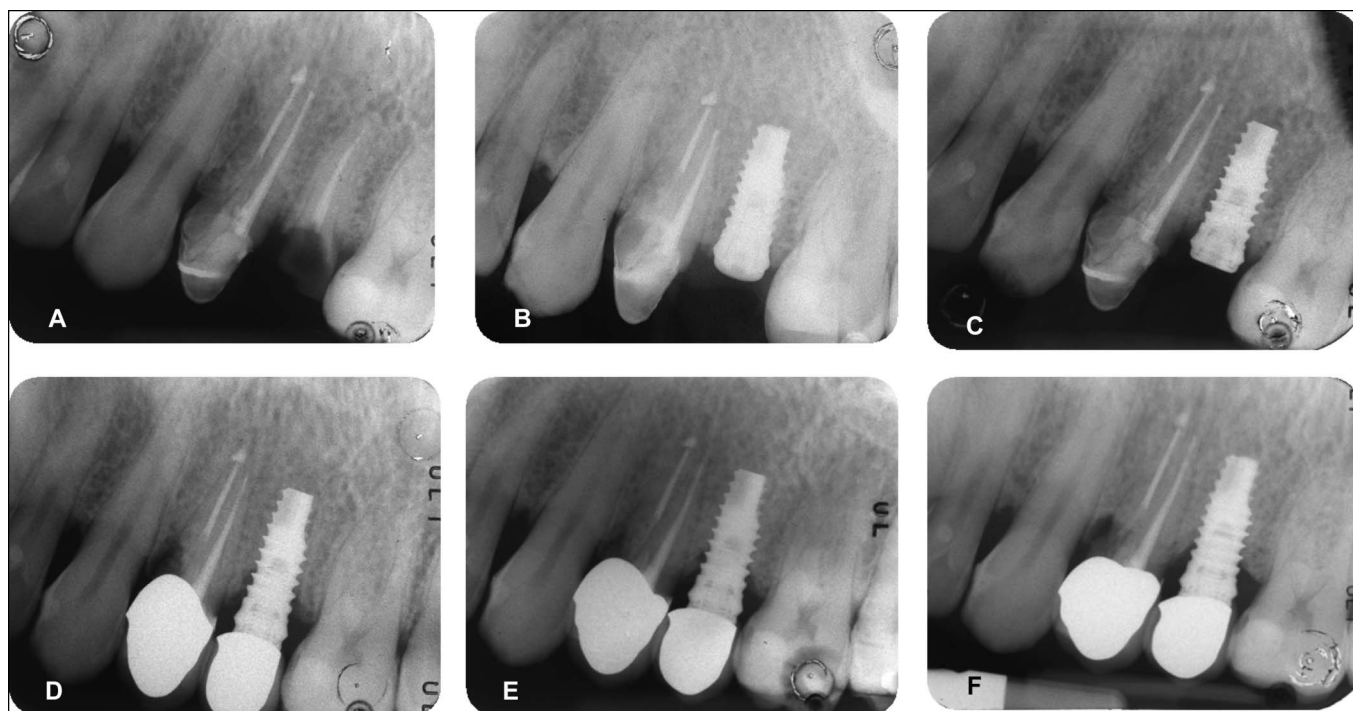


Fig. 2. A 52-year-old, nonsmoker male patient with a fractured 2.5. **A**, Preoperative rx. Tooth was extracted and after an adequate debridement of the alveolar socket, a 4.25 × 10-mm implant was inserted immediately (**B**). No complications were observed postoperatively. At 3 months, periapical rx (**C**) and impression were taken and a metal ceramic crown was cemented after approximately 4 months. In this patient, a marked MBL was detected after 6 months (**D**), that is 3 months after loading but remained stable at 12- (**E**) and 24-month (**F**) evaluations.

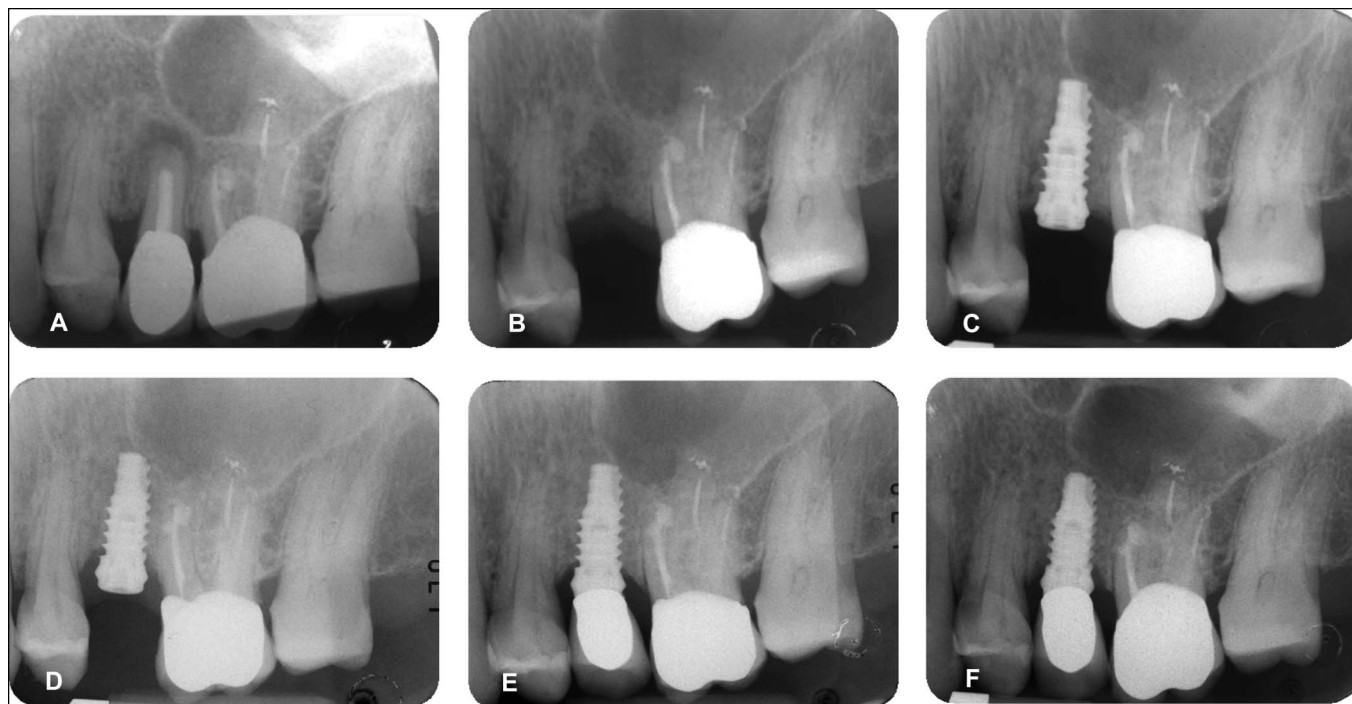


Fig. 3. A 56-year-old, nonsmoker female patient with a previously apicoexptomized 2.5 and apical abscess; tooth presented a vertical root fracture and extraction was performed. Pre-extraction rx (A). Two months after extraction, an adequate bone healing was observed (B) and a 3.8×10 -mm implant was inserted with a flapless approach (C). After 3 months, periapical rx (D) and impression were taken, and a metal-ceramic crown was cemented on the implant abutment approximately 1 month later. A modest MBL was observed after 3 months (preloading) and was stable after 6 months (not reported), 12 (E), and 24 (F) months.

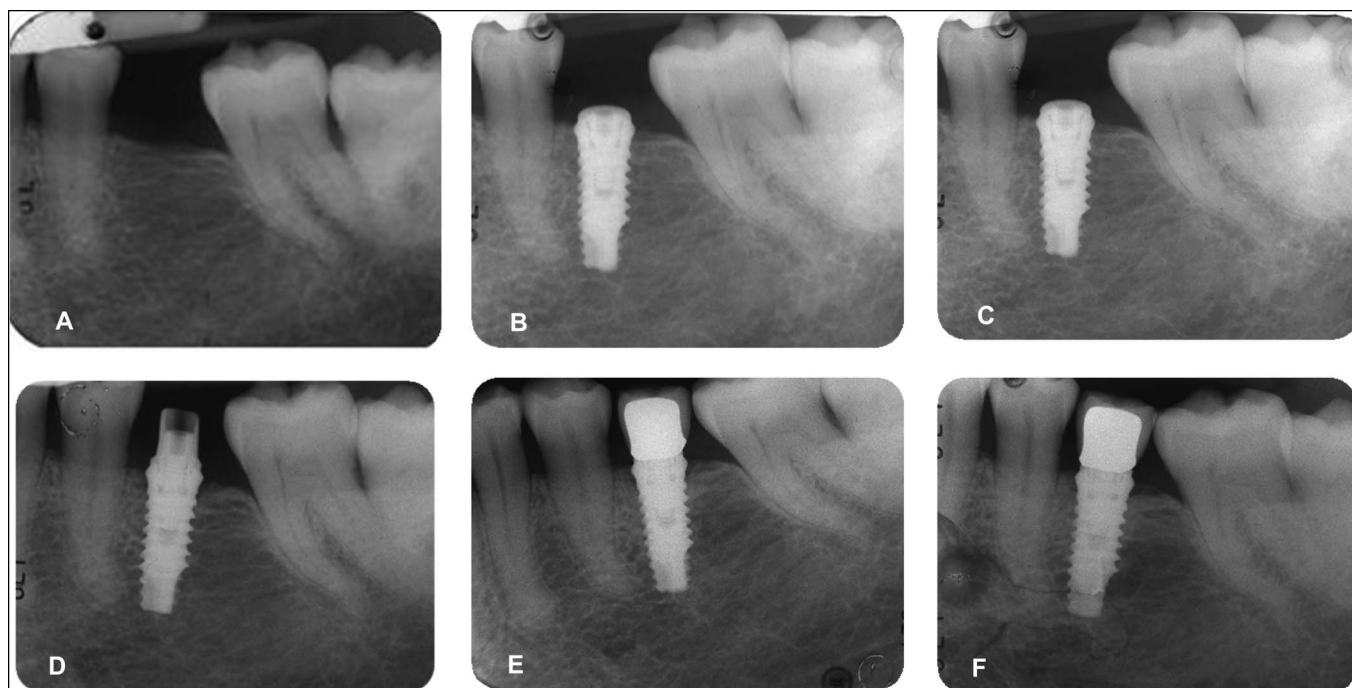


Fig. 4. (From left to right) A 49-year-old, nonsmoker female patient with a 3.6 edentulous area (A): Tooth was extracted 12 months before. A 3.8×10 -mm premium implant was inserted with flapless technique (B). No complications were observed. At 1 (C) and 3 months (D), follow-up bone levels were stable. A metal-ceramic crown was cemented on the implant abutment approximately 2 weeks later. MBL remained stable at 6 months (periapical rx not reported), 12 (E), and at 24 months (F) of follow-up.

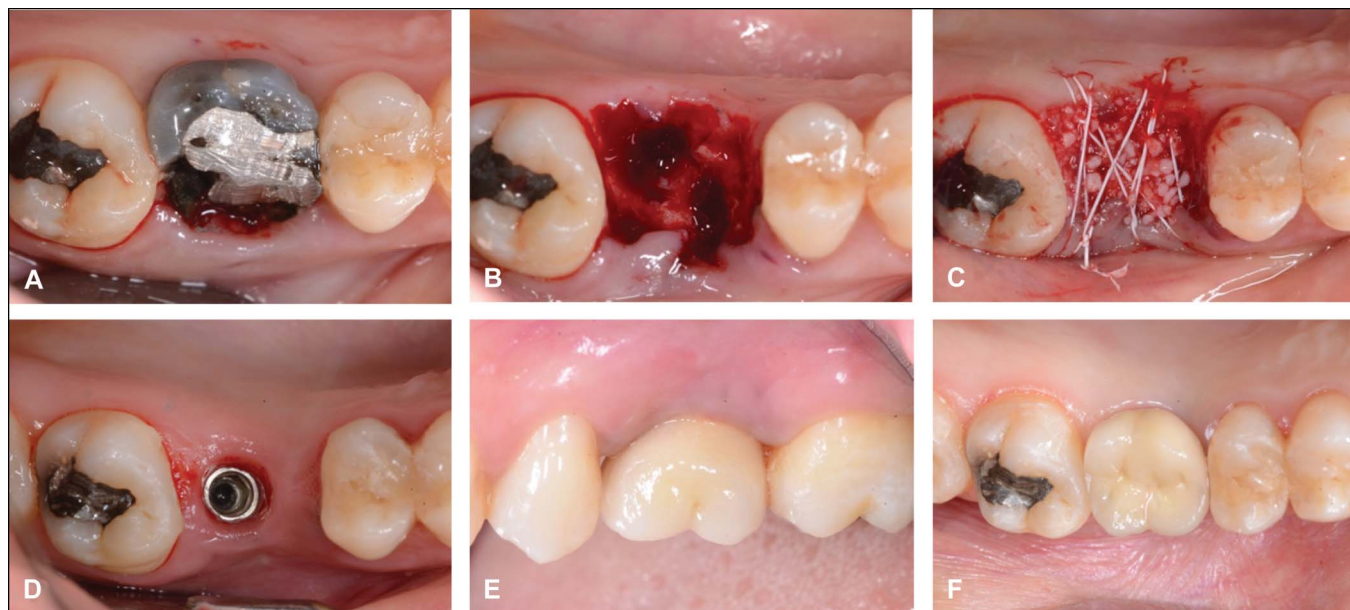


Fig. 5. Immediate implant placement clinical photograph sequence of a fractured “hopeless” upper molar; tooth presented no clinical or radiological signs of acute infection. Pre-ex (A), postextraction (B), after implant placement (premium 5.0 × 10 mm), and bone graft insertion (C). D, Soft tissues at the impression (3 months after implant placement). Definitive metal-ceramic crown, vestibular (E), and occlusal (F) view. Note the soft tissue growth and the gingival adaptation of the crown with no signs of inflammation.

All x-rays were scanned with a slide scanner with a resolution of 968 dpi and a magnification factor of ×20. The known length and diameter of implants were used to calibrate the measurement.

The crestal marginal bone and the bone-implant interface were examined to evaluate the marginal bone morphology. MBL was assessed at the mesial and distal implant surfaces by measuring the distance between the reference point of the implant platform to the most

coronal bone-to-implant contact level using a scale divided into 0.1 mm steps according to previous studies^{29,30} and corrected according to the know height and width of each implant.³

A single-blind radiographic evaluation was performed by 1 additional examiner. Before evaluating the radiographs, the examiner was calibrated by using well-defined instructions and reference radiographs with different marginal bone level measures.

Statistical Analysis of the MBL

Statistical analysis was performed using SPSS Statistics (Software version, 23.0). Descriptive statistics were computed as mean ± SDs for quantitative variables. Frequencies and percentages were used for qualitative variables. Marginal bone level variations were evaluated by measuring MBL. The inferential analysis was performed by creating a multilevel linear regression model at 2 levels (patient and implant). Moreover, the multilevel model investigated factors associated with MBL. The covariates used at the patient level were “gender” (females/males) and “age” (years). At implant level, covariates used were “implant placement group” (immediate, early/delayed) and “location” (maxilla/mandible); at time level the covariates used were T₁ (1 month), T₃ (3 months), T₆ (6 months), T₁₂ (12 months), and T₂₄ (24 months). α-level was previously set at 0.05.

Comparisons of MBL values among T₁, T₃, T₆, T₁₂, and T₂₄ for each covariate used at the implant level (implant placement group and location) were performed by means of nonparametric Friedman test and Wilcoxon test, given the nongaussian MBL distribution (Kolmogorov–Smirnov

Table 3. Multilevel Analysis Exploring Factors Associated With MBL

Parameter	Coefficient	SE	P	95% CI*
Intercept	1.221	0.666	0.67	−0.087 to 2.530
Age	−0.002	0.408	0.997	−0.803 to 0.800
Sex (reference: male)				
Females	0.230	0.578	0.690	−0.906 to 1.366
Implant placement group (reference: delayed)				
Immediate	−0.129	0.076	0.088	−0.278 to 0.019
Early	−0.400	0.073	0.0001	−0.543 to −0.257
Implant location (reference: mandible)				
Maxilla	−0.200	0.061	0.001	−0.320 to −0.079
MBL evaluation time (reference: T ₂₄ [24 mo])				
T ₁ (1 mo)	−0.684	0.102	0.0001	−0.885 to −0.482
T ₃ (3 mo)	−0.621	0.103	0.0001	−0.824 to −0.419
T ₆ (6 mo)	−0.425	0.103	0.0001	−0.627 to −0.222
T ₁₂ (12 mo)	−0.191	0.109	0.08	−0.405 to 0.023

A significant influence of implant placement group (immediate, early, and delayed group) and location (mandible/maxilla) on MBL was reported. Values in bold are statistically significant ($P < 0.05$).

*CI = confidence interval.

Table 4. MBL (Mean $\pm$ SD) for Implant Placement Group

	Preloading T ₁	T ₃	T ₆	Postloading T ₁₂	T ₂₄
Immediate (n = 29)	0.10 $\pm$ 0.36 ^{a,A}	0.15 $\pm$ 0.35 ^{a,A}	0.24 $\pm$ 0.35 ^{a,A}	0.60 $\pm$ 0.69 ^{ab,B}	0.78 $\pm$ 0.70 ^{b,B}
Early (n = 29)	0.01 $\pm$ 0.19 ^{a,A}	0.04 $\pm$ 0.29 ^{a,A}	0.14 $\pm$ 0.47 ^{a,A}	0.32 $\pm$ 0.64 ^{a,B}	0.48 $\pm$ 0.72 ^{a,B}
Delayed (n = 73)	0.27 $\pm$ 0.41 ^{b,A}	0.35 $\pm$ 0.62 ^{b,A}	0.67 $\pm$ 0.64 ^{b,B}	0.80 $\pm$ 0.84 ^{b,C}	1.02 $\pm$ 1.01 ^{b,C}

All groups showed MBL at all observational times. Delayed group showed greater MBL than the other 2 groups ($P < 0.02$). Comparisons between each time/group are performed by means of Wilcoxon test for paired data/Kruskal-Wallis analysis. In the vertical row, equal superscript small letters represent no statistical differences between times of placement ($P \geq 0.02$). In the horizontal row, equal superscript capital letters represent no statistical differences among times ($P \geq 0.005$).

test: $P \leq 0.005$). Multiple comparison tests were used by applying Bonferroni correction ($\alpha = 0.005$).

Comparisons of MBL values of the covariates used at the implant level for each time were performed using nonparametric Kruskal-Wallis analysis of variance (implant placement group) or Mann-Whitney U test (location); Bonferroni correction for multiple comparisons was applied ($\alpha = 0.02$).

RESULTS

According to the inclusion/exclusion criteria, 85 patients (131 implants) were studied with a mean age of 54.6 ± 11.8 years (46 women and 39 men). Eight patients were identified as smokers, consuming between 10 to 20 cigarettes/d and included in the study; these patients were distributed evenly across the 3 groups (3 in immediate, 2 in early, and 3 in delayed group).

Table 2 reports the number of implants for each group; 18 of 131 implants were placed in 8 light-smoker patients: (24.13% immediate, 6.89% early, 12.32% delayed).

The survival rate was 100% at all times. One nonsmoker patient from the early group dropped out after 6 months and was excluded from the study. Total drop out rate was 1.26%.

No wound infection, osteitis, bone graft sequestration, or implant loosening occurred during the follow-up period.

Mucositis was observed in 1 patient (early group) after 3 months caused by a recurrent unscrewing of the implant abutment. The abutment was removed, and the area was carefully treated with chlorhexidine 0.12%. After 1 month, a new abutment was screwed and a new metal-ceramic crown was cemented. No other mucositis was detected.

A representative series of periapical radiograph for the 3 groups is reported in Figures 2–4.

A clinical photograph sequence of an example of immediate implant is reported in Figure 5.

Multilevel analysis (Table 3) evidences the significant influence of implant placement group (immediate, early, and delayed groups) and location (mandible/maxilla).

MBL increased significantly with time (from 1-month and 3-month preloading evaluation to the 24-month postloading evaluation) in all the 3 groups (immediate, early, and delayed). The estimated difference in MBL of immediate versus delayed was 0.129 mm ($P = 0.088$) and early versus delayed was 0.400 mm ($P = 0.0001$).

The estimated differences between T₁, T₃, T₆, and T₁₂ versus T₂₄ were 0.68 mm ($P = 0.0001$), 0.62 mm ($P = 0.0001$), 0.43 mm ($P = 0.0001$), and 0.19 mm ($P = 0.08$), respectively.

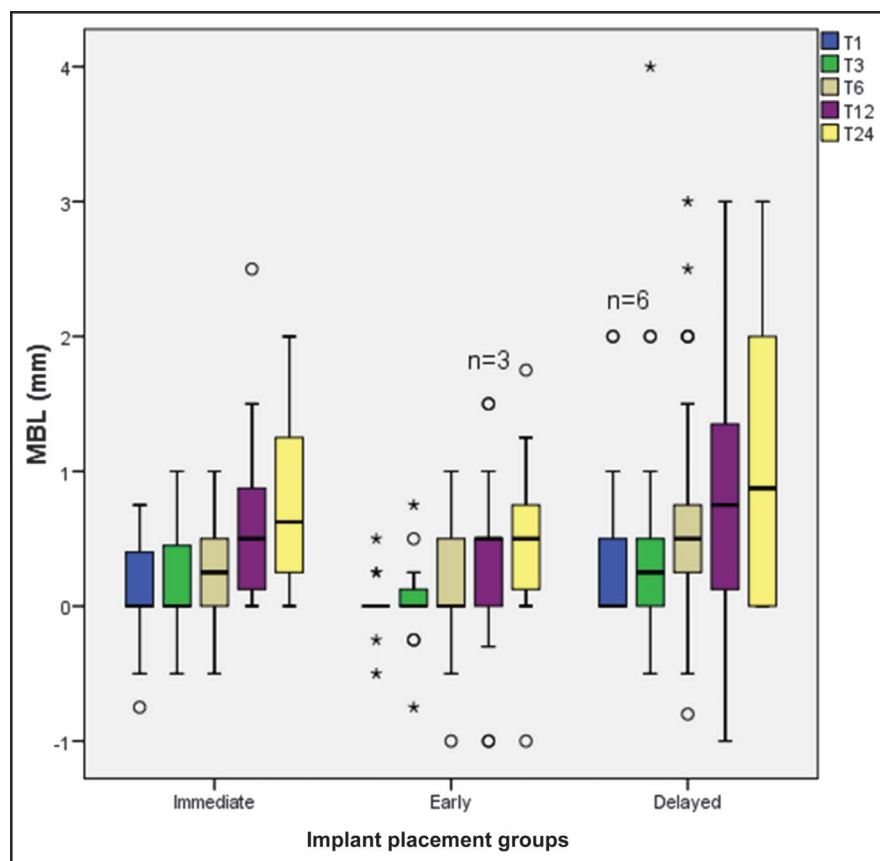


Fig. 6. Box plot representation of MBL (mm) at 1, 3, 6, 12, and 24 months from implant insertion. Early group showed the most reduced values of MBL and the most restricted range of bone loss in particular at 3 and 24 months. Delayed group showed the higher dispersion of MBL values and the greater range of bone loss. Rings indicate outlier cases with values between 1.5 and 3 times the interquartile range, that is, beyond the whiskers. Asterisks indicate extreme cases with values more than 3 times the interquartile range.

Table 5. MBL (Mean $\pm$ SD) for Location

	Preloading T ₁	T ₃	T ₆	Postloading T ₁₂	T ₂₄
Mandible (n = 54)	0.27 $\pm$ 0.39 ^{a,A}	0.30 $\pm$ 0.41 ^{a,A}	0.65 $\pm$ 0.66 ^{a,B}	1.00 $\pm$ 0.86 ^{a,C}	1.24 $\pm$ 0.98 ^{a,C}
Maxilla (n = 77)	0.13 $\pm$ 0.36 ^{a,A}	0.22 $\pm$ 0.60 ^{a,A}	0.32 $\pm$ 0.51 ^{b,AB}	0.41 $\pm$ 0.63 ^{b,B}	0.56 $\pm$ 0.76 ^{b,B}

Mandibular sites presented greater MBL values than maxilla sites at all the observational times ($P < 0.05$ after 6 months).

Comparisons between each time/group are performed by means of Wilcoxon test for paired data/Mann-Whitney U test.

In the vertical row, equal superscript small letters represent no statistical differences between locations ($P \geq 0.05$).

In the horizontal row, equal superscript capital letters represent no statistical differences among times ($P \geq 0.005$).

The estimated difference in MBL of maxilla versus mandible was 0.20 mm ($P = 0.001$).

Table 4 shows MBL distribution in the 3 implant placement groups. Early group showed the least MBL values at all observation times (1–24 months) (see also Box plot representation in Fig. 6). Moreover, the immediate group presented reduced MBL values when compared with the delayed group (Table 4 and Box plot in Fig. 6). Preloading MBL evaluation (T₁ and T₃) showed a statistically difference from postloading MBL in the delayed group after 6 months (T₆), and after 12 months in the immediate and early groups (T₁₂).

MBL values observed at the maxilla level were different from MBL values at the mandible level (Table 5). These values are statistically significant after 6, 12, and 24 months. Comparisons of MBL among the evaluation times (T₁–T₂₄) are statistically significant both for maxilla and mandible (Friedman test $P = 0.0001$).

DISCUSSION

This study used the clinical pre-extractive diagnosis as the main criteria for implant placement. For this reason, timing of implant placement (immediate, early, and delayed) was intentionally not determined randomly, as the purpose of the study was to use rigorous best clinical treatment criteria.

The presence of fistula, buccal bone deficit/defect, and pus connected to acute periapical abscess was considered and critically evaluated to establish a safe clinical procedure and guarantee the best clinical practice. Important bone destruction due to the presence of pathogenic bacteria is frequently present at the apical root level when an acute endodontic periapical lesion/abscess is present,^{19,20} increasing the risks of implant integration.^{22,24}

One recent systematic review concluded that there is limited evidence that immediate implant placement in sites affected by acute endodontic lesions leads to clinical outcomes comparable with those of immediate implants in healthy sites.²¹ Unfortunately, in many investigations the classification of endodontic infection is often vague and not well defined, according to other systematic reviews.^{14,15} For example, it is clinically evident and extremely important to distinguish a chronic periapical lesion with a modest radiolucency at the root apex from a large bone destruction/resorption with pus and fistula (frequently in the presence of pain). AAE Consensus Conference³¹ well described the clinical conditions, but it is already evident that “at chair side” in many cases, it is difficult to define the level of inflammation and when it is convenient or possible to place implant immediately after the root extraction. In our Endodontic Department, we opt for a strict clinical definition of acute periapical lesion. When swelling, tenderness, pus, and pain are present (in Latin “tumor, rubor, dolor, and calor”), we opt for the extraction and implant placement is postponed after a short period (2 months, early group implant placement).

In this study, early implants (type 2 for ITI classification) presented less periimplant marginal bone resorption at 1 and 3 months (preloading time) and the most reduced values at 6, 12, and 24 months compared with the other groups. Therefore, the data support the concept that 2 to 3 months after extraction is the ideal time for implant placement, obtaining a more than adequate periimplant bone level free from infection/bacteria with stable marginal bone morphology at 24 months. It does not seem necessary to wait to obtain a more structured bone.

In the early group, the tapered enlarged configuration of implant neck,

the flapless technique,³² and the transmucosal placement may have played a positive role, reducing bone stress and inducing a fast remodeling of marginal cortical bone.

Corticocancellous porcine bone was applied to fill socket gaps in immediate implant group (type 1 for ITI classification). This biological condition is favorable to new bone formation, as demonstrated by MBL values, with a reduced or modest bone resorption and with new bone formation. In all samples, neither pocket depth nor gingival bleeding was observed during the clinical controls. Other studies with other type of implants^{29,33–35} have demonstrated the low failure rate and the reduced periimplant MBL of immediate (postextractive) technique choice. It is possible to speculate that the high vascularization in healing area such as in immediately and early implant may be responsible for a fast bone formation that results in a reduced MBL.⁵

Delayed groups (type 4 for ITI Classification) showed a statistically significant higher MBL (compared with early implant group) and during the entire observation period. This study suggests that bone resorption around the neck of delayed implants is higher when a mature cortical bone is present, as in delayed group. These data are supported by many *in vivo* studies on animals, where histological observation offers evidence for modest new bone formation and great resorption of old/mature bone.^{36–39} Drilling procedures at the implant site may be responsible for bone necrosis and bone smear layer formation. These conditions induce the activation of osteoclasts and vascularization damage, conditions responsible for higher bone resorption of the mature cortical bone.^{36–39} It is probable that bone density around the implant decreases during the first weeks after implant insertion, whereas bone

remineralization requires more time to be activated, as a recent study demonstrated.³⁴

Periimplant marginal bone levels before loading (preloading) is rarely reported in other studies.⁴⁰ In this study (Table 4), preloading time was responsible for a relatively consistent MBL only in the delayed group.

The use of transmucosal/not non-submerged technique prevents a second re-entry surgery and probably may help to reduce MBL further at 3 to 6 months. Transmucosal implant placement is a treatment option that has been demonstrated to offer similar results^{41,42} to the submerged technique/implant placement and prevents many of its disadvantages such as second re-entry surgery, pain, healing time after exposure of implant and costs.

The implants used in this study presented a tulip-shaped switching platform emergence,⁴³ as visible in the periapical radiographs. This study demonstrates that implants with a Zir-Ti-treated surface and this platform switching design showed minimal MBL at 3 to 24 months within the limits defined by Albrektsson et al.⁴⁴

Implants placed in the maxilla showed less MBL compared with implants placed in the mandible. The tulip-shaped platform-switch configuration may be responsible for this optimal result in the maxilla and may have played some positive effect on remodeling perimarginal bone growth and guiding bone healing in immediate and early groups of the maxilla.

ZirTi surface was recently tested in an animal model by Mainetti et al^{36,37} in postextractive sockets and in control healed bone. It displayed optimal results after 1 and 3 months. Buccal bone loss was less pronounced in postextractive immediate implants than in the control group. After 3 months, no differences were observed between the 2 groups.^{36,37}

Galindo-Moreno et al² have demonstrated that an MBL of 0.44 mm or more in the first 6 months is an indication of fast periimplant bone loss progression. Therefore, in the delayed group, we may speculate a higher MBL progression during time. Further

evaluation at longer observation periods may confirm these data. However, we may presume that in a long-term study, other conditions play important roles, such as periodontal bacteria contamination, loading stress, abutment-implant connection and location, etc. For example, in a long-term (10 years) study, Schropp et al⁴⁵ demonstrated that timing of implant application (early, delayed, and late) did not significantly affect MBL in time.

CONCLUSION

The implant placement in sites with recent previous endodontic affected lesion does not represent a critical factor and is a safe procedure responsible for limited bone loss. Delayed implant placement demonstrated greater risks of MBL.

DISCLOSURE

The authors claim to have no financial interest, either directly or indirectly, in the products or information listed in the article.

APPROVAL

Ethical approval was not required in this study as no experimental procedures were performed.

ROLES/CONTRIBUTIONS

BY AUTHORS

C. Prati performed all the surgical treatment, clinically designed the study, and wrote the manuscript; C. Pirani performed the prosthetic rehabilitation; F. Zamparini contributed to prosthetic procedures and made the radiographic assessment; M. Rosaria Gatto performed statistical analysis; M.G. Gandolfi designed the clinical study; and A. Piattelli critically revised the manuscript.

REFERENCES

1. Quirynen M, Vogels R, Alsaadi R, et al. Predisposing conditions for retrograde peri-implantitis and treatment suggestions. *Clin Oral Implants Res.* 2005;16:599–608.

2. Galindo-Moreno P, León-Cano A, Ortega-Oller I, et al. Marginal bone loss as success criterion in implant dentistry: Beyond 2 mm. *Clin Oral Implants Res.* 2015;26:e28–e34.

3. Galindo-Moreno P, León-Cano A, Ortega-Oller I, et al. Prosthetic abutment height is a key factor in peri-implant marginal bone loss. *J Dent Res.* 2014;93:80s–85s.

4. Monje A, Suarez F, Galindo-Moreno P et al. A systematic review on marginal bone loss around short dental implants (<10 mm) for implant-supported fixed prostheses. *Clin Oral Implants Res.* 2014;25:1119–1124.

5. Qian J, Wennerberg A, Albrektsson T. Reason for marginal bone loss around oral implants. *Clin Implant Dent Relat Res.* 2012;14:792–807.

6. Lincoln-Bell C, Diehl D, Bell BM, et al. The immediate placement of dental implant into extraction site with periapical lesions: A retrospective chart review. *J Oral Maxillofac Surg.* 2011;69:1623–1627.

7. Lopez-Martinez F, Moreno GG, Olivares-Ponce P, et al. Implants failures related to endodontic treatment. An observational retrospective study. *Clin Oral Implants Res.* 2015;26:992–995.

8. Renvert S, Quirynen M. Risk indicators for peri-implantitis. A narrative review. *Clin Oral Implants Res.* 2015;26:15–44.

9. Zhao D, Wu Y, Xu C, et al. Immediate dental implant placement into infected vs. non-infected sockets: A meta-analysis. *Clin Oral Implants Res.* 2016;27:1290–1296.

10. Romanos GE, Froum S, Costa-Martins S, et al. Implant periapical lesions: Etiology and treatment options. *J Oral Implantol.* 2011;37:53–63.

11. Chrcanovic BR, Albrektsson T, Wennerberg A. Dental implants inserted in fresh extraction sockets versus healed sites: A systematic review and meta-analysis. *J Dent.* 2015;43:16–41.

12. Montoya-Salazar V, Castillo-Oyagüe R, Torres-Sánchez C, et al. Outcome of single immediate implants placed in post-extraction infected and non-infected sites, restored with cemented crowns: A 3-year prospective study. *J Dent.* 2014;42:645–652.

13. Hämmerle CH, Chen ST, Wilson TG Jr. Consensus statements and recommended clinical procedures regarding the placement of implants in extraction sockets. *Int J Oral Maxillofac Implants.* 2004;19:26–28.

14. Waasdorp JA, Evian CI, Mandracchia M. Immediate placement of implants into infected sites: A systematic review of the literature. *J Periodontol.* 2010;81:801–808.

15. Chrcanovic BR, Martins MD, Wennerberg A. Immediate placement of implants into infected sites: A systematic review. *Clin Implant Dent Relat Res*. 2015;17:e1–e6.
16. Nemcovsky CE, Artzi Z, Moses O, et al. Healing of dehiscence defects at delayed-immediate implant sites primarily closed by a rotated palatal flap following extraction. *Int J Oral Maxillofac Implants*. 2000;15:550–558.
17. Corbella S, Taschieri S, Samaranayake L, et al. Implant treatment choice after extraction of a vertically fractured tooth. A proposal for a clinical classification of bony defects based on a systematic review of literature. *Clin Oral Implants Res*. 2014;25:946–956.
18. Crespi R, Capparre P, Gherlone E. Fresh socket implants in periapical infected sites in humans. *J Clin Periodontol*. 2010;81:920–926.
19. Foschi F, Cavrini F, Montebugnoli L, et al. Detection of bacteria in endodontic samples by polymerase chain reaction assays and association with defined clinical signs in Italian patients. *Oral Microbiol Immunol*. 2005;20:289–295.
20. Cavrini F, Pirani C, Foschi F, et al. Detection of *Treponema denticola* in root canal systems in primary and secondary endodontic infections. A correlation with clinical symptoms. *New Microbiol*. 2008;31:67–73.
21. Lindeboom JA, Tjiook Y, Kroon FH. Immediate placement of implant in periapical infected sites: A prospective randomized study in 50 patients. *Oral Surg Oral Med Oral Pathol Oral Radiol Endodontol*. 2006;101:705–710.
22. Lee CT, Chuang SK, Stoupel J. Survival analysis and other clinical outcomes of immediate implant placement in sites with periapical lesions: Systematic review. *Int J Oral Maxillofac Implants*. 2015;30:268–278.
23. Ørstavik D, Kerekes K, Eriksen HM. The periapical index: A scoring system for radiographic assessment of apical periodontitis. *Endod Dent Traumatol*. 1986;2:20–34.
24. Lang NP, Lui P, Lau KY, et al. A systematic review on survival and success rates of implants placed immediately into fresh extraction sockets after at least 1 year. *Clin Oral Implants Res*. 2012;23:39–66.
25. Taxel P, Ortiz D, Shafer D, et al. The relationship between implant stability and bone health markers in postmenopausal women with bisphosphonate exposure. *Clin Oral Invest*. 2014;18:49–57.
26. World Medical Association Declaration of Helsinki. Ethical principles for medical research involving human subjects. *JAMA*. 2013;310:2191–2194.
27. Vandembroucke JP, von Elm E, Altman DG, et al. Strengthening the reporting of observational studies in epidemiology (STROBE): Explanation and elaboration. *PLoS Med*. 2007;4:1628–1654.
28. Dodson TB. A guide for preparing a patient-oriented research manuscript. *Oral Surg Oral Med Oral Pathol Radiol Endod*. 2007;104:307–315.
29. Sanz M, Cecchinato D, Ferrus J, et al. Implants placed in fresh extraction sockets in the maxilla: Clinical and radiographic outcomes from a 3-year follow-up examination. *Clin Oral Implants Res*. 2014;25:321–327.
30. Thoma DS, Sanz Martin I, Benic GI, et al. Prospective randomized controlled study comparing two dental implant systems: Demographic and radiographic results at one year of loading. *Clin Oral Implants Res*. 2014;25:142–149.
31. Gutmann JL, Baumgartner JC, Gluskin AH, et al. Identify and define all diagnostic terms for periapical/periradicular health and disease states. *J Endod*. 2009;35:1658–1674.
32. Prati C, Zamparini F, Scialabba VS, et al. A 3-year prospective cohort study on 132 calcium phosphate blasted implants: Flap vs flapless technique. *Int J Oral Maxillofac Implants*. 2016;31:413–423.
33. Covani U, Canullo L, Toti P, et al. Tissue stability of implants placed in fresh extraction sockets: A 5-year prospective single-cohort study. *J Periodontol*. 2014;85:323–332.
34. Barone A, Toti P, Quaranta A, et al. The clinical outcomes of immediate versus delayed restoration procedures on immediate implants: A comparative cohort study for single-tooth replacement. *Clin Implant Dent Relat Res*. 2014;17:1114–1126.
35. Crespi R, Cappare P, Polizzi E, et al. Fresh-socket implants of different collar length: Clinical evaluation in the aesthetic zone. *Clin Implant Dent Relat Res*. 2015;17:871–878.
36. Mainetti T, Lang NP, Bengazi F, et al. Immediate loading of implants installed in a healed alveolar bony ridge or immediately after tooth extraction: An experimental study in dogs. *Clin Oral Implants Res*. 2015;26:435–441.
37. Mainetti T, Lang NP, Bengazi F, et al. Sequential healing at implants installed immediately into extraction sockets. An experimental study in dogs. *Clin Oral Implants Res*. 2016;27:130–138.
38. Baffone GM, Botticelli D, Pantani F, et al. Influence of various implant platform configurations on peri-implant tissue dimensions: An experimental study in dog. *Clin Oral Implants Res*. 2011;22:438–444.
39. Baffone GM, Botticelli D, Pereira FP, et al. Influence of buccal bony crest width on marginal dimensions of peri-implant hard and soft tissues after implant installation an experimental study in dogs. *Clin Oral Implants Res*. 2013;24:250–254.
40. Gholami H, Mericske-Stem R, Kessler-Liechti G, et al. Radiographic bone level changes of implant-supported restorations in edentulous and partially dentate patients: 5-year results. *Int J Oral Maxillofac Implants*. 2014;29:898–904.
41. Sanz M, Ivanoff CJ, Weingart D, et al. Clinical and radiologic outcomes after submerged and transmucosal implant placement with two-piece implants in the anterior maxilla and mandible: 3-year results of a randomized controlled clinical trial. *Clin Implant Dent Relat Res*. 2015;17:234–246.
42. Sanz Martin I, Benic GI, Hämmerle CH, et al. Prospective randomized controlled clinical study comparing two dental implant types: Volumetric soft tissue changes at 1 year of loading. *Clin Oral Implants Res*. 2016;27:406–411.
43. Lazzara RJ, Porter SS. Platform switching: A new concept in implant dentistry for controlling post-restorative crestal bone levels. *Int J Periodontics Restorative Dent*. 2006;26:9–17.
44. Albrektsson T, Zarb G, Worthington P, et al. The long-term efficacy of currently used dental implants: A review and proposed criteria of success. *Int J Maxillofac Implants*. 1986;1:11–25.
45. Schropp L, Wenzel A, Spin-Neto R, et al. Fate of the buccal bone at implants placed early, delayed, or late after tooth extraction analyzed by cone beam CT: 10-year results from a randomized, controlled, clinical study. *Clin Oral Implants Res*. 2015;26:492–500.